Stabilizers, extensions and the enumeration of finite geometries

Human Compression project

Abstract

A classification by group actions requires both an equivalence relation and an admissibility test. We give a common way to extend a partial configuration, impose a support condition, and count its orbits using its actual stabilizer. We also prove recognition and exhaustive augmentation for coset geometries with an arbitrary chamber stabilizer. The applications concern commuting involutions, Suzuki polyhedra and the residually weakly primitive geometries of the Hall–Janko group. Their local group structure and exact computations remain explicit inputs.

1 The information a classification must retain

Two commuting involutions do not yet specify a regular polyhedron. A third involution must generate the prescribed group with them and satisfy all subgroup intersections. Likewise, panel subgroups do not specify a flag-transitive geometry until incident cosets have common representatives. The method developed here keeps these tests separate from the orbit reduction.

After fixing a partial configuration, its stabilizer acts on the remaining choices. Theorem UD.AdmissibleExtensions counts precisely the admissible choices, including a prescribed field of definition when present. It combines the elementary extension-orbit correspondence with the classical fixed-point count and Möbius inversion. Its role is to expose which hypotheses each application must prove; no priority is claimed for these classical principles. The finite-action theorem also records when a permutation image is faithful, so that an abstract automorphism group is not confused with a normalizer in one chosen action.

For coset geometries, the common paper contains the complete recognition and augmentation proofs. The Hall–Janko application supplies the subgroup data, the actual stabilizer actions and the finite outputs. For Suzuki groups, normalization and a subfield-position argument identify the support condition with generation. For symmetric and alternating groups, the transitive four-group actions supply the multiplicities and parity conditions. These local arguments are part of their applications rather than assumptions hidden inside the common theorem.

All groups in the counting statements are finite. The extension-orbit correspondence and the elementary coset realization are stated without that restriction. A geometry has a finite type set; incidence is reflexive and symmetric and two elements of the same type are incident only when equal. A flag is a pairwise incident set, a chamber has every type, and every flag must extend to a chamber. Firmness means that a flag missing one type extends in at least two ways. Residual connectedness concerns the incidence graph of every residue of rank at least two. The group action is type preserving; flag transitivity means transitivity on flags of every prescribed type.

A C-group marking is a generating family of distinct nonidentity involutions $(r_i)_{i\in I}$ such that, with $G_J = \langle r_i : i \in J\rangle$ and $G_\varnothing = 1$,

$$G_J \cap G_K = G_{J\cap K} \qquad (J, K \subseteq I).$$

For $I = \{0, \dots, d-1\}$ it is a string marking if $r_i r_j = r_j r_i$ whenever $|i - j| \geq 2$. The applications retain the standard correspondence between string C-group markings and regular abstract polytopes. Reversing the ordered markers corresponds to duality. A rank-three marking is therefore a regular polyhedron only after generation and all intersections have been established; endpoint orders alone do not give this conclusion.

2 Finite actions and extensions

Theorem UD.Orbits (orbit data and normalizers). Let a finite group A act on a finite set Ω, and choose one representative H of each conjugacy class of subgroups that occurs as a point stabilizer. Let m_H be the number of orbits isomorphic to A/H, with $m_H = 0$ for all omitted classes. Then the numbers m_H determine the A-set up to equivariant isomorphism, and

$$\ker(A \curvearrowright \Omega) = \bigcap_{m_H>0} \operatorname{core}_A(H). \tag{5.3}$$

Its equivariant permutation group is abstractly

$$C_{\mathrm{Sym}(\Omega)}(A) \cong \prod_H \big((N_A(H)/H) \wr \mathrm{Sym}(m_H)\big). \tag{5.4}$$

If the action $\rho : A \hookrightarrow \mathrm{Sym}(\Omega)$ is faithful, conjugation gives an exact sequence

$$1 \longrightarrow C_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow N_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow \{\alpha \in \mathrm{Aut}(A) : m_{\alpha(H)} = m_H \text{ for all } H\} \longrightarrow 1. \tag{5.5}$$

No splitting of this sequence is asserted.

Proof. Every orbit is equivariantly A/H, where H is the stabilizer of a selected point, and two such transitive sets are isomorphic exactly when the stabilizers are conjugate. This proves the classification by multiplicities and the kernel formula.

An equivariant permutation preserves the isomorphism type of each orbit. For one orbit A/H, an equivariant map is determined by the image of H. That image nH must be fixed by H, and bijectivity makes its stabilizer equal to H; hence $n \in N_A(H)$. Conversely each such n gives a permutation $aH \mapsto anH$. Composition corresponds to multiplication in the opposite order, so inversion identifies this permutation group with $N_A(H)/H$. For m_H copies, one may independently choose such a map in every copy and permute the copies, yielding the wreath product. Different isomorphism types are independent, proving (5.4).

Faithfulness makes conjugation by a normalizing permutation τ induce a unique automorphism α of A, with

$$\tau\rho(a)\tau^{-1} = \rho(\alpha(a)).$$

If a point has stabilizer H, its image under τ has stabilizer $\alpha(H)$; therefore the multiplicities must be invariant. The kernel of this conjugation map is the centralizer. Conversely, suppose α preserves the multiplicities. Pair each orbit with a target orbit whose stabilizer class is $\alpha(H)$, and choose base points x, y with actual stabilizers $H, \alpha(H)$, adjusting within the target orbit if needed. Define $\tau(ax) = \alpha(a)y$. The stabilizer equality makes this well-defined and bijective on that pair of orbits, and the paired definitions combine to a permutation of Ω. It satisfies the displayed intertwining equation, proving surjectivity and exactness. □

Proposition UD.ExtensionOrbits (classification above a retained base). Let a group B act on sets Z, E, and let $p : Z \to E$ be equivariant. For one representative e of each orbit in $B\backslash E$, there is a natural bijection

$$B\backslash Z \;\cong\; \coprod_{e \in B\backslash E} B_e\backslash p^{-1}(e). \tag{5.6}$$

Proof. Translate any $z \in Z$ so that $p(z)$ is the selected representative of its orbit. If two translated points over that representative are B-equivalent, an element taking one to the other fixes their image e, so belongs to B_e. Conversely a B_e-equivalence is already a B-equivalence. These two observations prove surjectivity and injectivity of (5.6). □

3 Admissible extensions with prescribed support

Let a finite group A act on finite sets Z and E, and let $p : Z \to E$ be equivariant. Let $R \subseteq Z$ be invariant; it is the set passing the application's admissibility tests. Suppose a finite poset L with greatest element 1 has an A-action by order automorphisms and an equivariant map $s : R \to L$. Write $R_l = \{z \in R : s(z) \le l\}$. The individual sets R_l need not be invariant. Choose one e in each A-orbit in $p(R)$, and put $R(e) = R \cap p^{-1}(e)$.

Theorem UD.AdmissibleExtensions. The equivalence classes of admissible extensions of exact support 1 are in bijection with the disjoint union of the A_e-orbits on $\{z \in R(e) : s(z) = 1\}$. Their number is

$$\sum_{e \in A\backslash p(R)} \frac{1}{|A_e|} \sum_{a \in A_e} \sum_{l \in L} \mu(l,1) |\operatorname{Fix}_{R(e)}(a) \cap R_l|,$$

where μ is the Möbius function of L.

Proof. The greatest element is fixed by every order automorphism, so the exact support condition is invariant. Apply UD.ExtensionOrbits to the restriction of p to that subset. Its fiber equivalence is exactly the A_e-action: an element carrying one extension over e to another necessarily fixes e. An empty exact-support fiber contributes zero, so representatives may be taken from all of $p(R)$.

For each $z \in R$, the defining inverse identity of the incidence-algebra Möbius function gives

$$\sum_{s(z) \le l \le 1} \mu(l,1) = \begin{cases} 1, & s(z) = 1, \\ 0, & s(z) < 1. \end{cases}$$

Sum over the points of $R(e)$ fixed by a particular $a \in A_e$. This gives the inner sum in the formula. Finally, for a finite group H acting on a finite set Y, count the pairs (h, y) with $hy = y$. Their number is both $\sum_h |\operatorname{Fix}_Y(h)|$ and $\sum_y |H_y|$. Each orbit contributes $|H|$ to the second sum by the orbit–stabilizer identity. Division by $|H|$ gives the number of orbits. Use this identity with $H = A_e$ and the exact-support fiber, and sum over the representatives e. □

Corollary UD.ExactSupportOrbits. If A acts on a finite set X with an equivariant support map to L, the number of exact-top-support orbits is

$$\frac{1}{|A|} \sum_{a \in A} \sum_{l \in L} \mu(l,1) |\operatorname{Fix}_X(a) \cap \{x : s(x) \le l\}|.$$

Proof. Take E to be a one-point set and $R = Z = X$. □

Taking a one-element support poset instead recovers counting in ordinary extension fibers. For an involution D on a finite set Y, it gives $(|Y| + |\operatorname{Fix}_Y(D)|)/2$ orbits. A division by two is therefore justified without a correction term only after freeness has been proved.

Lemma UD.OddOrbits. Let a finite group A of odd order act on X. Let D be a fixed-point-free involution of X commuting with this action. Then D has no fixed orbit in $A\backslash X$.

Proof. If $Dx = ax$, commutation and $D^2 = 1$ give $a^2 x = x$. The order of a is odd, so a is a power of a^2 and fixes x. Then $Dx = x$, a contradiction. □

For the divisor poset of an integer N, support often means the least field $\mathbb{F}_{2^d}$ containing all coordinates, where $d \mid N$. The application must establish that its subfield sets are exactly the support sublevel sets. If $A(d)$ counts data defined over that field, the number of data of exact degree N is

$$E(N) = \sum_{d \mid N} \mu(N/d) A(d).$$

If Frobenius acts freely on them with orbit length N, division by N counts its orbits. For odd N, a commuting fixed-point-free involution then permits another division by two, by UD.OddOrbits. These are conditional consequences of the common theorem, not substitutes for the generation and stabilizer calculations in the applications.

4 Cosets and flags

For subgroups $P_i \leq G$, typed coset Helly means that every pairwise intersecting family of right cosets $P_i g_i$, with distinct indices, has a common element. This property is separate from IP. It supplies a common representative for an incident flag, whereas IP determines the stabilizer of a flag already represented in this way.

Lemma UD.Helly (anchored criterion). A finite indexed subgroup family $(P_i)_{i\in I}$ has typed coset Helly if and only if the following condition holds. For every $i \in I$, every $J \subseteq I \setminus \{i\}$ with $|J| \geq 2$, and any selected $j_0 \in J$, each $g \in P_{j_0}$ satisfying $g \in P_i P_j$ for all $j \in J$ admits $k \in \bigcap_{j\in J} P_j$ with $P_i g = P_i k$.

Proof. The cosets $P_i g$ and P_j intersect precisely when $g \in P_i P_j$. Thus Helly gives a common element k, proving necessity. Conversely, induct on the number of cosets in a pairwise intersecting family. A family of at most two already has a common element. For a larger family, the induction hypothesis gives a common element of all but one coset. Right multiplication by its inverse turns those cosets into the base cosets P_j. The remaining coset meets P_{j_0}, so it has a representative $g \in P_{j_0}$. Its intersections with the other base cosets are precisely the product conditions in the statement. The criterion supplies k, and undoing the right translation gives a common element of the original family. □

5 Panel systems with a nontrivial chamber stabilizer

Lemma UD.CosetRealization. Let a group G act type-preservingly and flag-transitively on an incidence geometry of finite rank, and let $(x_i)_{i\in I}$ be a base chamber. Write $H_i = \mathrm{Stab}_G(x_i)$ and use a right action. Then

$$H_i g \longmapsto x_i^g$$

is a type-preserving G-equivariant isomorphism from the subgroup coset system to the given geometry.

Proof. Transitivity on one-object flags makes the map onto each type. It is well defined and injective because $x_i^g = x_i^h$ holds exactly when $gh^{-1} \in H_i$, which is equivalent to $H_i g = H_i h$. If two cosets share a representative k, their images are x_i^k, x_j^k and are incident, as the base objects are incident. Conversely, if two images are incident, transitivity on flags of their two types supplies a k carrying the base pair to them. The injective object correspondence then writes both cosets with that common representative. The map therefore preserves and reflects incidence. □

Let G be finite, let I be a finite type set, and let each panel subgroup P_i contain B. Put $K_\emptyset = B$ and $K_S = \langle P_i : i \in S\rangle$ for nonempty S, with $K_I = G$. This marking by subgroups is different from a marking by individual involutions.

Theorem UD.PanelRecognition. Suppose

$$K_S \cap K_T = K_{S\cap T} \quad (S, T \subseteq I), \qquad [P_i : B] \geq 2, \tag{E4}$$

and suppose the maximal parabolics $H_i = K_{I\setminus\{i\}}$ have typed coset Helly. Then their coset incidence system is a firm flag-transitive residually connected geometry with chamber stabilizer B. Conversely every finite-rank firm flag-transitive residually connected coset geometry is obtained this way by taking the panel stabilizers.

Proof. Equation (E4) identifies the stabilizer of every partial base flag with the corresponding K_S. UD.Helly supplies a common group element in the cosets of every flag, and hence flag transitivity and identification of the residues with the corresponding coset systems in K_S. Each rank-one residue has $[P_i : B]$ elements, giving firmness. A rank-two residue is connected because its two panel subgroups generate $K_{\{i,j\}}$. For larger rank, all panel subgroups occur in proper deletions, so the residue group is generated by its maximal parabolics. The usual coset-path argument proves connectedness: paths multiply successive elements of those subgroups, and conversely their products give paths.

Conversely let K_S stabilize the complementary base flag. Stabilizer intersections give (E4). In a connected residue, the subgroup generated by its maximal parabolics has exactly the orbit reachable by

coset paths, so it is the whole residue group. Induction on $|S|$ reconstructs K_S from the panels, starting with $K_\emptyset = B$. Flag transitivity gives typed Helly by moving a flag to the base flag. □

The source's remaining axioms are separate, exact local tests. RWPRI means that for every nonempty S there is an $i \in S$ with $K_{S\setminus\{i\}}$ maximal proper in K_S. Local two-transitivity means that P_i has two orbits on ordered pairs in $B\backslash P_i$; equivalently B has two orbits on that coset set. The source's geometric $(\mathrm{IP})_2$ means that every rank-two residue is a partial linear space or a generalized digon. It is not the subgroup identity (E4).

For the last test let P, Q be two panels, $P \cap Q = B$ and $K = \langle P, Q\rangle$. The coset graph is a digon exactly when $|P||Q| = |K||B|$. Otherwise local two-transitivity makes a single common-neighbour count sufficient. Choose $x \in Q \setminus B$. Neighbours of P are the Q-cosets represented by a right transversal of B in P; neighbours of Px are their translates by x. The stabilizer Q is two-transitive on its incident P-type points, so all distinct points having a common Q-type neighbour have the same number of common neighbours. This number must be one. Pairs with no common neighbour already satisfy partial linearity. This argument also applies when a local action has a nontrivial kernel.

5.1 Complete augmentation through maximal residues

Lemma UD.Transport. Suppose representatives of all ambient G-conjugacy classes of subgroups K and all K-conjugacy classes of maximal subgroups $H_0 < K$ are given. For a specified H, choose any transporter c with $H_0^c = H$ when one exists. All maximal overgroups of H in that inclusion class are precisely $(K^c)^n$ with $n \in N_G(H)$.

Proof. Conjugate any maximal inclusion $H < L$ to a representative K of the class of L. Its image of H is K-conjugate to one of the listed H_0. Comparing the resulting transporter with c leaves an element normalizing H. Conversely each displayed conjugate contains H maximally. Taking the union over all inclusion classes gives every maximal overgroup. □

Let $\mathcal{P}(B)$ be the finite set of overgroups P of B such that B is maximal proper in P and P acts two-transitively on $B\backslash P$. UD.Transport computes this entire set by retaining only the relevant local inclusions. The normalizer $N_G(B)$ acts on $\mathcal{P}(B)$, and its permutation image has exactly the same configuration orbits. We retain the actual subgroups behind the permutation labels, so kernel reduction does not change any geometric test.

Theorem UD.MaximalAugmentation. Start with all rank-one panels modulo $N_G(B)$. For every accepted panel tuple, let H be its generated join. For every maximal overgroup K of H, try every $P \in \mathcal{P}(B)$ with $P \le K$ and $P \not\subseteq H$, modulo the pointwise stabilizer of the old panel tuple. Apply UD.PanelRecognition and all the local source tests, then remove duplicates under $N_G(B)$ and type permutations. Iterating until the next level is empty gives every target geometry with Borel B and acting group a subgroup of G.

Proof. For a target rank-r geometry, RWPRI supplies at least one maximal parabolic H in its acting group K. Deleting its complementary panel gives a rank-$(r-1)$ residue with the same Borel and all inherited source axioms. Induction puts its panel tuple in a previously enumerated orbit. Its remaining panel lies inside K and outside H, and UD.Transport includes K among the maximal overgroups to be tried. UD.ExtensionOrbits shows that taking pointwise-stabilizer representatives loses no extension; a smaller stabilizer than the full setwise one merely permits duplicates. Final orbit tests remove those duplicates. Conversely every accepted tuple satisfies exactly the recognition and local conditions.

By (E4) and proper panels, adjoining a new type strictly enlarges the generated join: if P_i lay in K_S for i outside S, then $P_i \cap K_S = B$ would force $P_i = B$. Hence every tuple gives a chain of strict subgroup inclusions from B. Its rank is at most $\Omega([G : B])$, the number of prime factors with multiplicity. Thus a run through the next level must reach an empty level. The certificate requires that emptiness, not merely a chosen numeric cap. Finally take B over all G-conjugacy classes of subgroups. □

Attribution and use

The orbit classification, centralizer description, fixed-point count and Möbius inversion are classical. The common extension correspondence and the arbitrary-chamber-stabilizer recognition and augmentation proofs are reused from the previously reviewed project proof bank (U.Orbits, U.ExtensionOrbits and the enumeration manuscript, Sections 2 and 5). The combined counting interface removes separate invocations of those principles; it does not establish new historical classification results. Coset geometry and the equivalence between string C-groups and abstract regular polytopes retain their established meanings; the latter correspondence is an external input in the polyhedron applications.

The full common paper is charged once within this topic. Its overlap with other topic theories does not reduce that charge. Complete local proofs, required programs and the separately identified diagnostics accompany the applications. No numerical compression factor is assigned by this edition.